Computer Assisted Proofs

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Introduction

Introduction

A big spectrum

	own computer programs
pencil and paper proofs	⋮
⋮	automatic theorems provers
formally verified proofs	⋮
	interactive theorems provers

Introduction

Some families in the spectrum

- Automatic theorems provers for first-order logic
- Satisfiability modulo theories solvers
- Inductive theorem provers
- Interactive proof assistants
- Tools for the verification of programs

Automatic Theorems Provers for First-Order Logic

Automatic Theorems Provers for First-Order Logic (ATPs)

Some features

- The TPTP World

  "A well know and established infrastructure that supports research, development, and deployment of automated theorem proving systems for classical logics." (Sutcliffe 2010, p. 1.)

  See `http://www.tptp.org/`.

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  The World Championship for Automated Theorem Proving!

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- Some ATPs for (classic) first-order logic: E, Equinox, Metis, SPASS and Vampire.

The TPTP Languages

History

- FOF (Full First-Order Form) (Sutcliffe 2009)
- CNF (Clause Normal Form) (Sutcliffe 2009)
- TFF (Typed First-Order Form)
  - TF0 (monomorphic) (Sutcliffe, Schulz, Claessen and Baumgartner 2012)
  - TF1 (rank-1 polymorphism) (Blanchette and Paskevich 2013)
- THF (Typed Higher-Order Form)
  - TH0 (monomorphic) (Sutcliffe and Benzmüller 2010)
  - TH1 (rank-1 polymorphism) (Kaliszyk, Sutcliffe and Rabe 2016)

(continued on next slide)

The TPTP Languages

History (continuation)

- TXF (Typed Extended First-Order Form)
  - TX0 (monomorphic) (Sutcliffe and Kotelnikov 2018)
  - TX1 (rank-1 polymorphism) (Sutcliffe and Kotelnikov 2018)
- DHF (Dependently Typed higher-order Form)
  (Ranalter, Kaliszyk, Rabe and Sutcliffe 2025)

FOF Annotated Formulae and Problems

Description

`fof(name, role, formula)` where:

`name` identifies the formula,

`formula` is a FOL-formula and

`role` $\in$ { `axiom`, `definition`, `hypothesis`, `conjecture` }.

Each TPTP problem contains one or more annotated formulae.

FOF Syntax

<table>
<tr><th>Logical constant</th><th>Symbol</th></tr>
<tr><td>True</td><td>$true</td></tr>
<tr><td>False</td><td>$false</td></tr>
<tr><td>Conjunction</td><td>&</td></tr>
<tr><td>Disjunction</td><td>|</td></tr>
<tr><td>Negation</td><td>~</td></tr>
<tr><td>Conditional</td><td>=></td></tr>
<tr><td>Biconditional</td><td><=></td></tr>
<tr><td>Universal quantifier</td><td>![X] : p(X)</td></tr>
<tr><td>Existential quantifier</td><td>?[X] : p(X)</td></tr>
</table>

Satisfiability Modulo Theories Solvers

Satisfiability Modulo Theories Solvers (SMT Solvers)

Satisfiability Modulo Theories

- The study of automatic methods for checking the satisfiability of first-order formulae with respect to some background theory is called Satisfiability Modulo Theories (SMT) (Barret, Sebastiani, Seshia and Tinelli 2009).

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- Examples of background theories:

  Theories of real numbers, integers, arrays, lists, records, bit-vectors,...

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SMT-LIB

The SMT library and standard.

Satisfiability Modulo Theories Solvers (SMT Solvers)

Some SMT solvers

- Z3 (Microsoft), CVC4 and veriT.

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- See `http://en.wikipedia.org/wiki/Satisfiability_Modulo_Theories`.

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Some SMT solvers

- Z3 (Microsoft), CVC4 and veriT.
- See http://en.wikipedia.org/wiki/Satisfiability_Modulo_Theories.

Z3 demo

See Z3 home page.

Inductive Theorem Provers

Inductive Theorem Provers (ITPs)

- There is not standard

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- There is not standard
- ITPs (Google for List of All Inductive Theorem Provers in The World).

Interactive Proof Assistants

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Description

"Proof assistants are computer systems that allow a user to do mathematics on a computer, but not so much the computing (numerical or symbolical) aspect of mathematics but the aspects of proving and defining. So a user can set up a mathematical theory, define properties and do logical reasoning with them."
(Geuvers 2009, p. 3.)

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Examples

- Based on set theory: Isabelle/ZFC, Metamath and Mizar
- Based on higher-order logic: HOL4, HOL Light and Isabelle/HOL
- Based on type theories: Agda, Rocq and Lean.

Some Proof Assistants

- Agda
- Isabelle
- Lean
- Rocq

Agda 2

- Chalmers University of Technology and University of Gothenburg (Sweden)
- Based on Martin-Löf Type Theory
- Direct manipulation of proofs-objects
- Back-ends to Haskell (GHC) and JavaScript
- Written in Haskell

Isabelle

- University of Cambridge (England) and Technical University of Munich (German)
- Based on higher order logic
- Tactic-based
- Extraction of programs to Haskell, OCaml, Scala and SML
- Written in SML
- Integration with ATPs and SMT solvers

Lean

- Microsoft Research and Carnegie Mellon University (USA)
- Based on dependent type theory
- Declarative style and tactic-based
- Written in C++
- Integration with ATPs and SMT solvers

Rocq

- INRIA (France)
- Based on the Calculus of Inductive Constructions
- Tactic-based
- Extraction of programs to Haskell, OCaml and Scheme
- Written in OCaml (with a bit of C)

Real Applications

Verification of Landau's *Grundlagen der Analysis*

1977 PhD thesis from Eindhoven University of Technology (Automath).
(van Benthem Jutting 1977)

Mizar Mathematical Library (MML)

One of the largest repository of formalised mathematics (1990–current).

MML includes (Bancerek, Byliński, Grabowski, Korniłowicz, Matuszewski, Naumowicz and Pak 2018):

- over 12000 definitions
- over 59000 theorems
- over 1290 articles
- over 253 authors

Proof of the Four Colour Theorem

1976 Every planar graph is four colorable. Announce in the Bulletin of AMS.
(K. Appel and Haken 1976)

1977 Description of the proof in Scientific American.
(K. Appel and Haken 1977)

1977 Simpler proof (still using a computer).
(Robertson, Sanders, Seymour and Thomas 1997)

2005 Formalised proof. Microsoft report (Rocq).
(Gonthier 2005)

2008 Description of the formalised proof in Notices of AMS.
(Gonthier 2008)

CompCert: A Verified Compiler for a Large Subset of C

2006 Journal (and proceedings) publication: POPL 2006 (Rocq).
(Leroy 2006)

2009 Journal publication.
(Leroy 2009a)

2009 Description in CACM.
(Leroy 2009b)

seL4: Formal Verification of an Operating System Kernel

2009 Proceedings of SOSP 2009 (Isabelle/HOL).

2010 Description in CACM.
(Klein, Andronick, Elphinstone, Heiser, Cock, Derrin, Elkaduwe, Engelhardt, Kolanski, Norrish, T. Sewell, Tuch and Winwood 2010).

Proof of the Odd-Order Theorem

1963 Solvability of groups of odd-order (Feit and Thompson 1963).

2012 Announce of the formalised proof (Rocq).
(Gonthier and Théry 2012)

2013 Formalised proof. Journal publication.
(Gonthier, Asperti, Avigad, Bertot, Cohen, Garillot, Le Roux, Mahboubi, O'Connor, Biha, Pasca, Rideau, Solovyev, Tassi and Théry 2013)

Proof of Kepler Conjecture

1998 Hales announced proof of Kepler Conjecture.

2005 Journal publication "uncertified" in Annals of Mathematics.
(Hales 2005)

2014 Announce of the formalised proof (Isabelle and Isabelle/HOL, HOL Light, OCaml, CamlP5 and GLPK).
(Hales 2014)

2017 Formalised proof. Journal publication.
(Hales, Adams, Bauer, Dang, Harrison, Hoang, Kaliszyk, Magron, McLaughlin, T. T. Nguyen, Q. T. Nguyen, Nipkow, Steven, Pleso, Rute, Solovyev, Ta, Tran, Trieu, Urban, Vu and Zumkeller 2017)

Verification of the Functional Behaviour of a Floating-Point Program

2014 Journal publication (Frama-C).
(Marché 2014)

CakeML: A Verified Implementation of ML

2014 Journal publication (HOL4).
(Kumar, Myreen, Norrish and Owens 2014)

Verification of a Cryptographic Primitive: SHA-256

2015 Journal publication (Rocq).
(A. W. Appel 2015)

Certifying a File System using Crash Hoare Logic

2015 Proceedings of SOSP 2015 (Rocq).

2017 Description in CACM.
(Chajed, Chen, Chlipala, Kaashoek, Zeldovich and Ziegler 2017)

Rigorous Test-Oracle Specification and Validation for TCP/IP and the Sockets API

2001 Technical report (HOL4).

2015 Proceedings publication (2008, 2006, 2005, 2002, 2001).

2017 Journal publication.
(Bishop, Fairbairn, Mehnert, Norrish, Ridge, P. Sewell, Smith and Wansbrough 2018)

Verified Compilation on a Verified Processor

2019 Proceedings of PLDI 2019 (HOL4).
(Lööw, Kumar, Tan, Myreen, Norrish, Abrahamsson and Fox 2019)

Proof-Checking Euclid Book I

2019 Journal publication (HOL Light and Rocq).
(Beeson, Narboux and Wiedijk 2019)

2022 Description in The American Mathematical Monthly.
(Beeson 2022)

2020 Proceedings of IJCAR 2020 (SMT solver).

2021 Description in CACM.
(Monroe 2021)

Verified Approximation Algorithms

2020 Proceedings of IJCAR 2020 (Isabelle/HOL).

2022 Journal publication.
(Eßmann, Nipkow, Robillard and Sulejmani 2022)

Brack: A Verified Compiler for Scheme via CakeML

2026 Proceedings of CPP 2026 (HOL4).
(Lasnier, Yallop and Myreen 2026)

Tools for the Verification of Programs

Dedicate Tools and Systems for the Verification of Functional Programs

- Hip: The Haskell Inductive Prover

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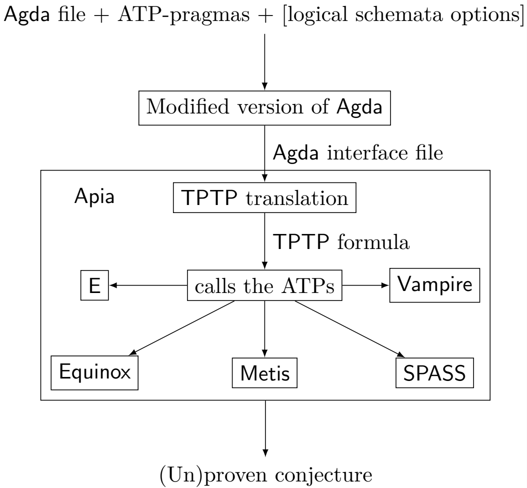
- Sparkle: An interactive proof assistant for Clean

Dedicate Tools and Systems for the Verification of Functional Programs

- Hip: The Haskell Inductive Prover
- Sparkle: An interactive proof assistant for Clean
- Zeno: An automated proof system for Haskell program properties

An Application: Reasoning about Functional Programs

Combining Agda with ATPs*



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